

A place to live

Every form of life—each species of plant, animal, or microbe—has a specific set of needs that means it can only thrive in suitable places.

Habitats are places where organisms live. A habitat can be as small as a rotting log or as big as the open ocean, but each one offers a different mixture of conditions that suits a particular community of species. There, the inhabitants that are adapted to these conditions—to the habitat's climate, food, and all other factors—can grow and survive long enough to produce the next generation.

Life between the tides

Nowhere can the diversity between habitats be seen better than where the land meets the sea. Conditions vary wildly on a rocky shore—from the submerged pools of the lower levels to the exposed land higher up. As the tide moves in and out daily, many species must be adapted to a life spent partly in the open air and partly underwater.

Lower shore

Life on the lowest part of the shore usually stays covered by seawater—a good habitat for organisms that cannot survive being exposed to the air.

Middle shore

On the middle zone of the shore, a seaweed called bladder wrack spends about 50 percent of its time in the water and 50 percent out of the water as the tide rises and falls.

High shore

Only the toughest ocean species survive on the highest, driest part of the shore. The seaweed here, called channelled wrack, can survive losing more than 60 percent of its water content.

Serrated wrack

Serrated wrack seaweed survives on the lower shore—where it is only uncovered when the tides are at their lowest.

Snails

Many animals, such as snails that graze on algae, can only feed when they are underwater.

